

[Optional] Lab: Create a Security Architecture Diagram for Jackson Corporation

Estimated time needed: 30 minutes

Overview

In this optional lab, you'll create a comprehensive security architecture diagram that visualizes your recommended solutions for Jackson Corporation.

Prerequisites:

- Completed "Create Your First Security Architecture Diagram" lab earlier in the course
- Completed "Final Project: Recommendations to Improve Network Security"

Objectives

After completing this lab, you will be able to:

1. Apply security architecture principles to a real-world scenario
2. Translate security recommendations into a visual architecture diagram
3. Design a multi-layered security infrastructure with appropriate zones and controls
4. Document security improvements in a clear, professional format

The challenge

Jackson Corporation has accepted your security recommendations from the previous assignment. Now they need to see how these improvements will fit together in an actual architecture. Your task is to create a security architecture diagram that shows:

- How the infrastructure will be restructured
- Where new security controls will be placed
- How traffic will flow through the secured environment
- Where critical assets will be protected

Exercise 1: Plan your architecture

Before you start drawing, spend 10 minutes planning your diagram on paper or in notes.

Task A: Identify your security zones

Based on the Jackson Corporation scenario, you'll need at least these zones:

► Hint: What zones should Jackson Corporation have?

Your task: List out the security zones you'll include in your diagram.

Task B: Map components to zones

For each of the seven areas from your recommendations, identify:

- What new components or security controls need to be added?
- Which security zone should each component be placed in?
- Why does that placement make sense?

► Hint: Common components to include

Your task: Create a simple table mapping each component to its appropriate zone.

Task C: Define traffic flows

Think about how different users and systems will communicate:

► Hint: Key traffic flows to consider

Your task: List the major traffic flows you need to show in your diagram.

Exercise 2: Create your security architecture diagram

Now it's time to create your diagram. Use any tool you're comfortable with (draw.io, Visio, Lucidchart, PowerPoint, PlantUML, or even pen and paper).

What to include

Your diagram **MUST** include:

1. Security zones (clearly labeled)

- Internet Zone (Untrusted)
- DMZ Zone (Semi-Trusted)
- Internal Zone (Trusted)

2. Security controls at boundaries

- External Firewall (between Internet and DMZ)
- Internal Firewall (between DMZ and Internal)
- VPN Gateway for remote access
- IDS/IPS for threat detection
- WAF for web application protection

3. Core infrastructure components

- Web Server (relocated to DMZ)
- Application Servers (in Internal zone)
- Database Servers (in Internal zone, protected)
- Monitoring/SIEM system (collecting logs from all zones)
- Employee workstations

4. Traffic flows

- Arrows showing how data moves between components
- Labels indicating protocols (HTTPS, VPN, internal APIs, etc.)
- Indication of encryption where applicable

5. Security annotations

- Notes explaining what each security control does
- Brief descriptions of OWASP protections (WAF, encryption, input validation)
- Any special security measures for sensitive data

6. Legend

- Explanation of colors, shapes, and symbols used

Guiding questions as you draw

Ask yourself these questions to ensure your diagram is complete:

About zone separation:

- Is the web server now in the DMZ instead of the internal network?
- Is the database protected by at least two layers of firewalls?
- Can external users reach the internal network directly? (They shouldn't!)

About security controls:

- Does every zone boundary have a firewall or security control?
- Is there a WAF protecting the web server from OWASP vulnerabilities?
- Is there an IDS/IPS monitoring for breaches?
- Is there a SIEM collecting logs from all systems?

About traffic flow:

- Can you trace the path of a customer request from the internet to the database?
- Does traffic pass through security controls at each step?
- Are all connections labeled with protocols and encryption status?

Reflection questions

After completing your diagram, consider these questions:

1. How did visualizing the architecture help you understand the security improvements better than just writing recommendations?
2. Which security control do you think will have the biggest impact on Jackson Corporation's security posture, and why is it visible in your diagram?
3. If you were presenting this diagram to Jackson Corporation's executives (non-technical), what would you emphasize?
4. What trade-offs or challenges might Jackson Corporation face when implementing this architecture?

Summary

Congratulations! You've now applied your security architecture skills to a real-world scenario. You've:

- Translated written security recommendations into a visual architecture
- Documented how security controls work together
- Created a professional diagram for stakeholder communication

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